

CHARLOTTE DOUGLAS, NC, USA

WMO: 723140

Lat: 35.224N

Lon: 80.955W

Elev: 728

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	24.8	1.5	6.1	30.3	5.9	7.7	31.5	22.6	53.1	18.9	48.2	5.0	310	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	94.2	74.6	91.9	74.0	89.8	73.5	77.4	88.3	76.4	86.8	75.4	85.2	6.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	132.2	81.1	73.5	127.9	80.2	72.6	124.3	79.6	41.1	88.6	40.2	87.0	39.3	85.5	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	16.1	13.6	13.8	97.1	4.2	3.1	10.7	99.3	8.3	101.1	5.9	102.8	2.9	105.0		
			11.6	78.9	3.8	1.3	8.9	79.8	6.7	80.6	4.6	81.3	1.9	82.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	42.0	45.8	52.7	61.4	69.2	76.6	79.5	78.5	72.6	61.9	51.4	45.1	(6)
(7)		DBStd	15.15	9.88	9.21	9.12	7.60	6.41	4.85	3.73	3.92	5.69	7.60	8.14	8.84	(7)
(8)		HDD50	843	287	176	79	8	0	0	0	0	6	80	207		(8)
(9)		HDD65	3030	714	541	392	156	36	2	0	11	148	412	618		(9)
(10)		CDD50	5031	38	58	162	349	595	798	913	885	679	377	122	55	(10)
(11)		CDD65	1742	0	2	10	47	166	350	449	420	240	53	4	1	(11)
(12)		CDH74	15729	2	11	98	451	1416	3244	4341	3863	1878	400	22	3	(12)
(13)		CDH80	6146	0	0	10	86	425	1342	1933	1623	648	78	1	0	(13)
(14)	Wind	WSAvg	6.1	6.3	6.9	7.4	7.4	6.5	5.7	5.3	5.2	5.8	5.4	5.6	5.7	(14)
(15)	Precipitation	PrecAvg	43.1	3.5	3.1	4.0	3.4	3.1	4.1	4.2	4.1	3.8	3.3	3.1	3.6	(15)
(16)		PrecMax	58.5	6.6	5.7	8.3	8.3	9.6	7.8	8.9	9.1	8.0	13.7	9.4	8.7	(16)
(17)		PrecMin	28.9	0.9	1.1	1.3	0.9	0.2	0.8	1.7	0.5	0.3	0.0	0.4	1.3	(17)
(18)		PrecStd	7.0	1.5	1.2	1.8	1.9	2.0	2.0	1.7	2.1	1.8	2.8	2.0	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.8	75.5	81.3	86.1	91.0	96.5	96.9	96.5	92.8	86.5	77.0	73.0	(19)
(20)		0.4%	MCWB	61.1	61.9	62.7	66.2	71.8	74.1	74.9	74.4	71.3	70.9	62.9	62.3	(20)
(21)		2%	DB	66.3	69.9	76.7	82.7	87.6	92.9	94.2	93.4	89.6	81.7	72.8	67.7	(21)
(22)		2%	MCWB	58.5	57.7	60.5	64.9	70.0	74.1	75.4	74.6	71.6	67.7	60.8	60.9	(22)
(23)		5%	DB	62.6	65.6	72.6	79.4	84.8	90.1	91.7	90.8	86.8	78.5	69.4	63.9	(23)
(24)		5%	MCWB	56.0	56.1	58.1	63.0	69.4	73.3	74.6	74.4	71.0	65.7	60.3	58.1	(24)
(25)		10%	DB	58.1	61.6	68.3	75.9	81.9	87.5	89.4	88.2	83.8	75.1	66.1	60.3	(25)
(26)		10%	MCWB	51.3	52.5	55.6	61.7	67.8	72.3	74.1	73.6	70.3	65.0	58.3	54.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	65.0	66.0	70.4	75.0	77.4	78.8	78.7	76.5	75.2	68.3	66.2	(27)
(28)		0.4%	MCDWB	67.5	71.1	75.9	80.2	84.9	89.7	91.2	89.2	84.9	80.1	70.5	69.4	(28)
(29)		2%	WB	60.9	61.8	63.2	68.0	72.8	76.1	77.4	77.4	75.0	71.7	65.1	62.9	(29)
(30)		2%	MCDWB	64.7	67.3	72.0	77.4	82.7	87.7	88.8	87.9	83.1	77.1	69.6	66.2	(30)
(31)		5%	WB	57.1	58.1	61.1	66.0	71.2	75.0	76.4	76.3	73.6	69.1	62.8	59.4	(31)
(32)		5%	MCDWB	61.7	63.6	68.8	74.9	81.0	85.9	87.3	86.0	80.8	75.0	67.3	63.0	(32)
(33)		10%	WB	52.4	53.9	58.8	64.0	69.6	73.8	75.5	75.2	72.4	66.7	59.5	55.2	(33)
(34)		10%	MCDWB	56.8	59.3	65.7	72.6	78.6	84.1	85.7	84.1	79.5	72.8	64.6	58.5	(34)
(35)	Mean Daily Temperature Range		MDBR	18.9	19.9	21.2	21.9	20.4	19.0	18.6	18.4	18.5	20.8	21.4	18.5	(35)
(36)		5% DB	MCDBR	22.8	24.3	25.6	25.3	22.8	22.3	21.6	21.3	22.4	23.4	24.8	21.3	(36)
(37)		5% DB	MCWBR	15.7	15.5	13.3	11.6	9.2	7.5	6.9	6.6	7.9	11.0	14.7	14.9	(37)
(38)		5% WB	MCDBR	19.7	20.8	21.5	20.5	20.1	19.4	19.0	18.4	17.6	17.5	18.4	17.8	(38)
(39)		5% WB	MCWBR	15.7	15.6	13.2	11.1	8.7	7.5	7.0	6.6	7.6	10.2	13.7	14.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.322	0.360	0.403	0.458	0.485	0.536	0.517	0.428	0.380	0.330	0.312	(40)	
(41)		taud	2.498	2.469	2.397	2.270	2.154	2.117	1.999	2.038	2.273	2.385	2.486	2.533	(41)	
(42)		Ebn at Noon	283	293	289	280	265	257	242	245	263	267	274	277	(42)	
(43)		Edh at Noon	26	30	36	43	48	50	56	53	40	33	27	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	837	1043	1381	1725	1915	2035	1903	1742	1479	1218	924	720	(44)	
(45)		RadStd	49	121	99	134	156	146	126	82	131	151	76	67	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+211	(46)
(47)	Regional (69 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+292	+198	(47)

Nomenclature: See separate page